Pedro Beirão

Software Engineer

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About me

I care about writing software that is maintainable and that people actually want to use, most of my projects are open source, and some have active communities around them.

Technical skills

Languages

C | C++ | Python | JavaScript | Ruby | SQL | Java | C# | Bash

Frameworks

Ruby on Rails | Qt | Unity | Godot | SDL | React

Infrastructure

Git | CI/CD | Docker | CMake

Markup

LaTeX | Typst | HTML

Other

Node.js | OpenGL | OpenMP | Websockets | CRDT | RESTful APIs | IEC 61499 | QGIS

Languages

- English
- Portuguese
- Spanish
- Dutch

Experience

Lead Maintainer @ DoomSpeedDemosArchive *2024 - Present*

I am the Lead Maintainer of all DSDA related projects, and the organization as a whole:

- dsda-doom: A modern but conservative DOOM source port with a focus on speedrunning.
  C, CMake, OpenGL, SDL, CI/CD, Code reviewing
- dsda-launcher: A GUI launcher for the needs of dsda-doom.
  C++, Qt, CI/CD, Crossplatform, UI development
- dsdarchive.com: The website that hosts DOOM's speedrunning scene.
  Ruby on Rails, SQL, API, Full-Stack, Server hosting
- ENDOOM Viewer: An online viewer for VGA text-based graphics.
  JavaScript, Binary data
- Part of Doom Cross Port Collab team: A collaborative effort for creating new standards and ensure better cross port development.
  Standards design, Future proofing, Collaboration

Researcher @ SYSTEC *2025 - 2026*

Researched a new architecture for a collaborative browser-native tool for IEC-61499 engineering.

- Created PTERO, a web-based IDE for IEC 61499 engineering.
  Web development, Docker, Node, Websockets, CRDT, IEC 61499
- Published the article "PTERO: A Collaborative Browser-Native Engineering Environment for IEC 61499 Distributed Automation".
  LaTeX, Research
- Contributed to the development of DINASORE, an IEC 61499 runtime.
  Python, Docker, IEC 61499

Author of Jet Set Billy *2022*

A recreation of the cult classic game for the ZX Spectrum, Jet Set Willy.
Java, UML, Unit tests

Author of pyopenglDoom *2021*

Doom renderer using PyOpenGL, an exercise on reading binary structured data.
Python, OpenGL, Binary data

Author of mindcuber-python *2021*

Solving a Rubik's cube with a ev3 Mindstorms, without needing a SD card.
Python, Robot

Participated in Game Jams

Participated both solo and in (international) team for 3 multi-day Game Jams.
C#, Unity, Godot, Execution under pressure

Education

University of Porto 2021 - 2026

Porto, Portugal

- Master's in Computer Science
- Bachelor's in Computer Science

University of Twente 2025

Enschede, Netherlands

- One semester of ERASMUS